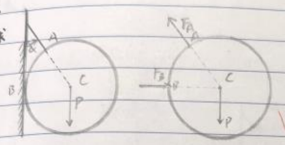
9.22

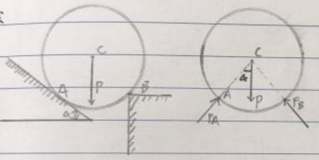
A+

1-2

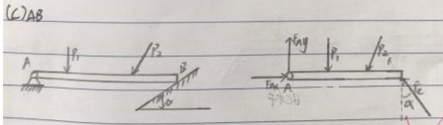
(a)球



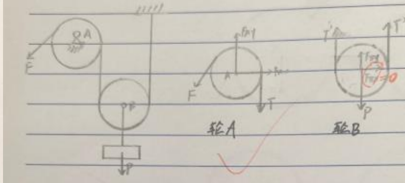
(b)球



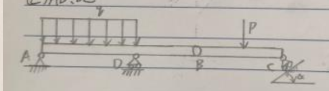
(C)AB

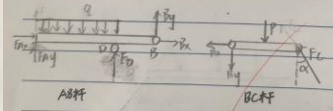


(d)轮A、轮B



(e)AB、BC

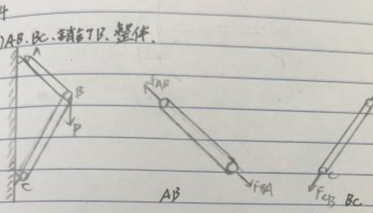


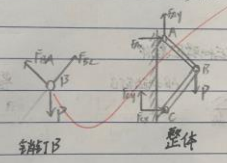


9.29

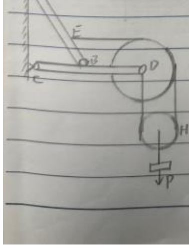
1-4

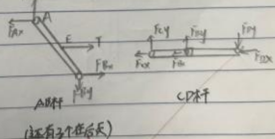
(a)AB、BC、销钉B、整体。





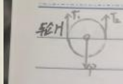
(b)AB、CD、轮H、销钉和轮D、整体



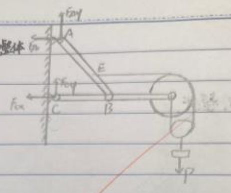


(注还有3个在后页)

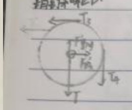
轮H



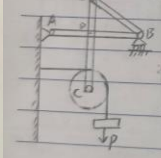
整体



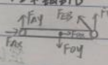
销钉和轮D.



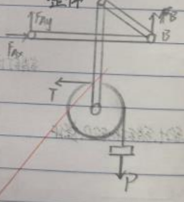
(C)AB和销钉B、CE和滑轮、整体



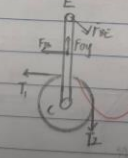
AB和销钉B

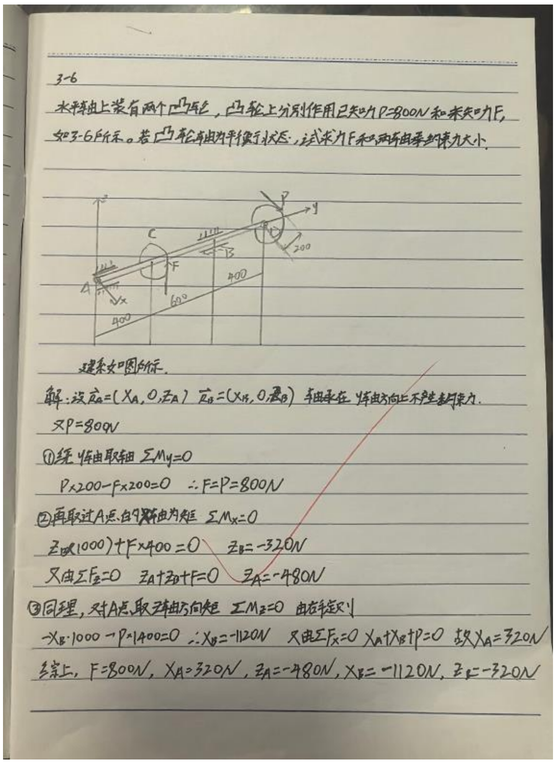


整体



CE和滑轮整体




A
C
B
F
P
x
y
400
600
400
200

3-6

水平轴上装有两个凸轮，凸轮上分别作用已知力P=800N和未知力F，如3-6所示。若凸轮轴为平衡状态，试求力F和两轴承约束力大小。

建系如图所示。

解：设 $\vec{F_A}=(X_A,0,Z_A)$ $\vec{F_B}=(X_B,0,Z_B)$ 轴承在y轴方向上不产生约束力。

又 $P=800N$

①绕y轴取轴 $\sum M_y=0$

$P\times200-F\times200=0$ $\therefore F=P=800N$

②再取过A点的~~x~~轴力矩 $\sum M_x=0$

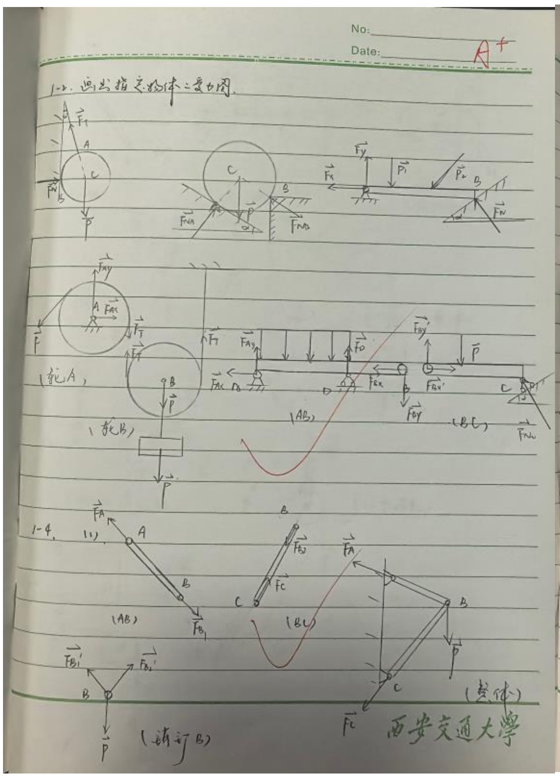
$Z_B(1000)+F\times400=0$ $Z_B=-320N$

又由 $\sum F_z=0$ $Z_A+Z_B+F=0$ $Z_A=-480N$

③同理，对A点取z轴方向矩 $\sum M_z=0$ 由右手定则

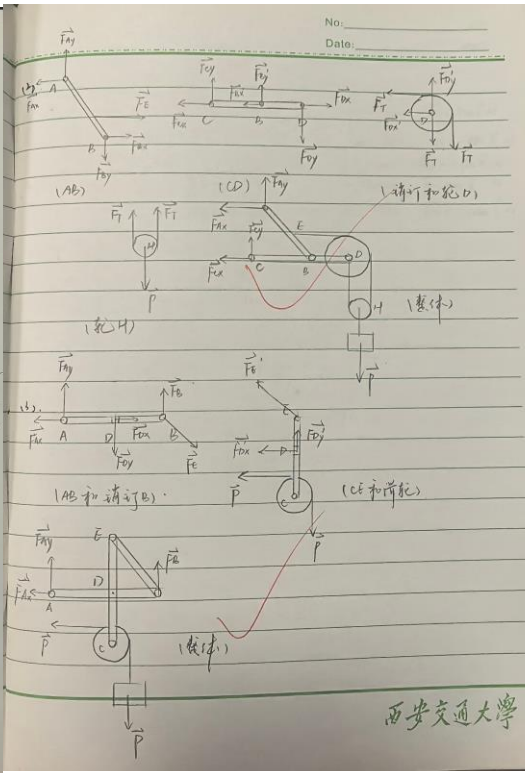
$-X_B\cdot1000-P\times1400=0$ $\therefore X_B=-1120N$ 又由 $\sum F_x=0$ $X_A+X_B+P=0$ 故 $X_A=320N$

综上，$F=800N$，$X_A=320N$，$Z_A=-480N$，$X_B=-1120N$，$Z_B=-320N$

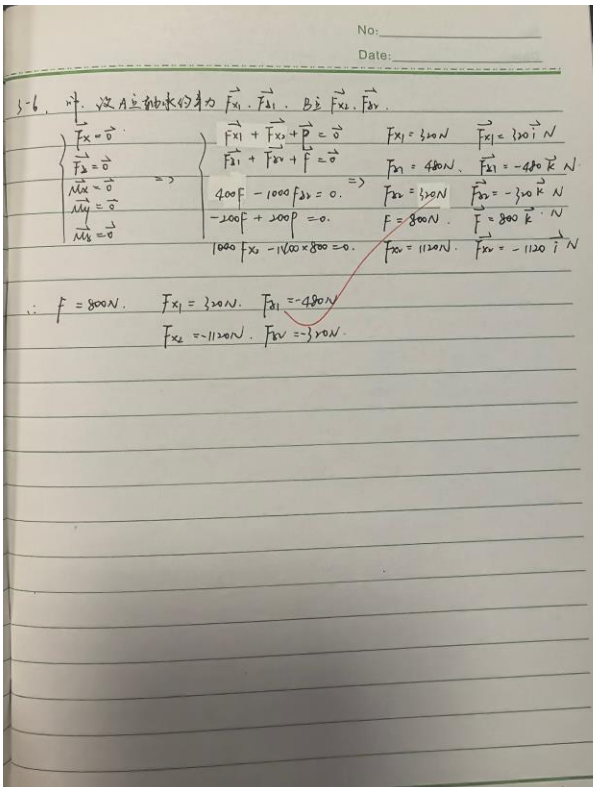


1-3. 画出指定物体之受力图.

1-4. (1).



(3).



No:

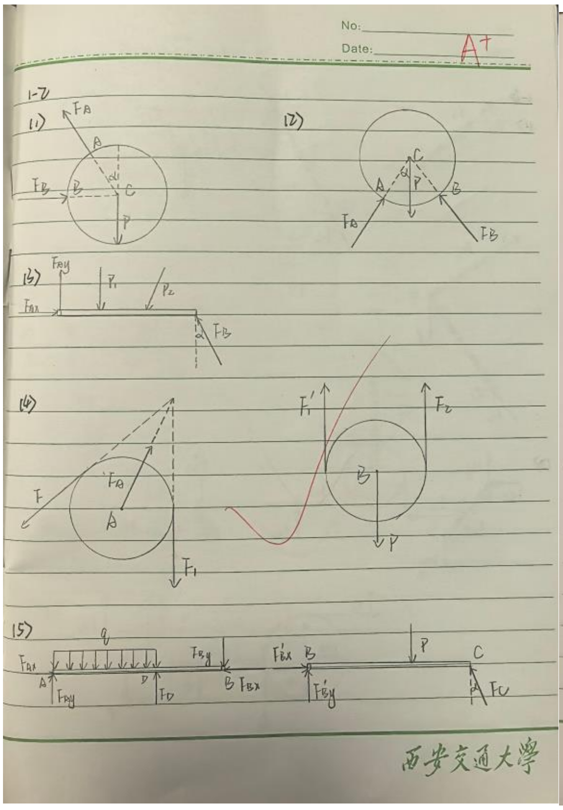
Date:

3-6、 解. 设A点轴承的约束力 $\vec{F}_{x1}$. $\vec{F}_{z1}$. B点 $\vec{F}_{x2}$. $\vec{F}_{z2}$.

$$\begin{cases}\vec{F}_x=\vec{0}\\ \vec{F}_z=\vec{0}\\ \vec{M}_x=\vec{0}\\ \vec{M}_y=\vec{0}\\ \vec{M}_z=\vec{0}\end{cases} \Rightarrow \begin{cases}\vec{F}_{x1}+\vec{F}_{x2}+\vec{P}=\vec{0}\\ \vec{F}_{z1}+\vec{F}_{z2}+\vec{F}=\vec{0}\\ 400F-1000F_{z2}=0.\\ -200F+200P=0.\\ 1000F_{x2}-1400\times800=0.\end{cases} \Rightarrow \begin{matrix} F_{x1}=320N & \vec{F}_{x1}=320\vec{i}\ N\\ F_{z1}=480N. & \vec{F}_{z1}=-480\vec{k}\ N\\ F_{z2}=320N & \vec{F}_{z2}=-320\vec{k}\ N\\ F=800N. & \vec{F}=800\vec{k}\ N\\ F_{x2}=1120N. & \vec{F}_{x2}=-1120\vec{i}\ N\end{matrix}$$

$\therefore$ $F=800N$. $F_{x1}=320N$. $F_{z1}=-480N$

$F_{x2}=-1120N$. $F_{z2}=-320N$.



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1-2

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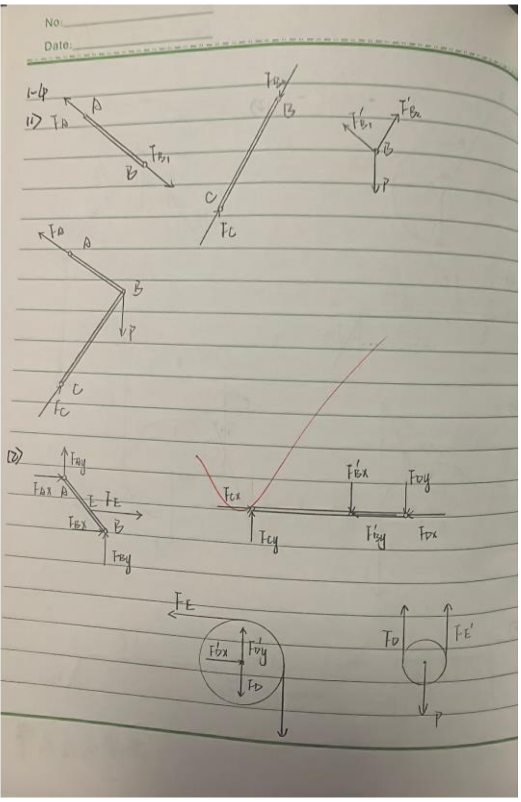
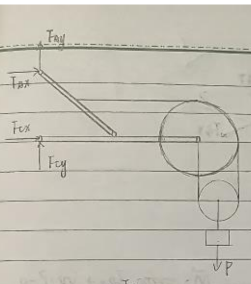


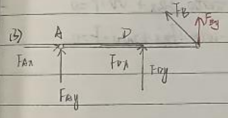
No:

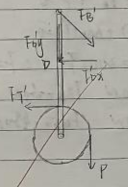
Date:

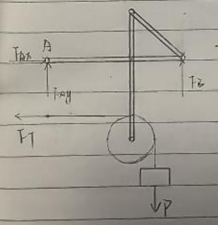
1-4



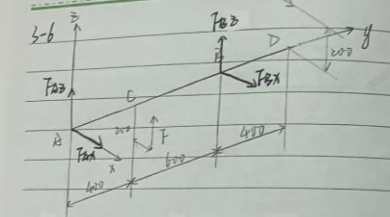
(3)







3-6



$$\begin{cases} F_x = F_{Ax} + F_{Bx} + P = 0 \\ F_z = F_{Az} + F_{Bz} + F = 0 \\ \vec{M_y} = M_D(\vec{P}) + M_C(\vec{F}) = \vec{0} \end{cases}$$

$M_z = -1000 \cdot F_{Bx} - 1400P = 0$

$M_x = 1000 \cdot F_{Bz} + 400 \cdot F = 0$

$\therefore 200 \times 800 = 200 \times F$ $\therefore F = 800N$

$\therefore F_{Bz} = -320N$ $F_{Bx} = -1120N$

$F_{Ax} = 320N$ $F_{Az} = -480N$